

SELVAKRISHNAN RAJENDRAN

+91 6380019373

selvakrish2949@gmail.com

[selvakrishnanrajendran](https://www.linkedin.com/in/selvakrishnanrajendran)

github.com/selvakrishnan

Education

Bannari Amman Institute of Technology, Sathyamangalam, Erode

Aug 2016 – May 2020

B.Tech Information Technology

CGPA: 7.29

Coursework: Data Science & Analytics, Cloud Computing, Data Structure & Algorithms

Work Experience

Data Engineer III

OCT 2024 – Present

Walmart Global Tech India (Industry: Retail)

Chennai, India

- Led the migration of ETL workflows from Automomic to Apache Airflow, improving scalability, observability, and reducing manual job failures.
- Designed and implemented DAGs (Directed Acyclic Graphs) in Airflow to automate complex data pipelines and ensure end-to-end data reliability.
- Developed and optimized ETL pipelines to identify and process blocking customer records, improving data quality and enabling timely business actions.
- Migrated Hive tables to Apache Hudi to support upserts, deletes, and incremental processing. Built Spark-based ingestion jobs, optimized Hudi table configurations, and improved query performance for downstream analytics teams.

Consultant (Data Engineer)

OCT 2022 – OCT 2024

Deloitte (Industry: Banking and Retail)

Chennai, India

- Planning, Migrating & Redesigning Entire Business Critical Data Pipelines from on-premises Oracle, MySQL, and MSSQL databases to Google BigQuery, facilitating enhanced data accessibility and scalability for advanced analytics.
- Architected and developed comprehensive dataflow pipelines for 58 distinct Source Files, encompassing rigorous data quality checks to ensure the accuracy and reliability of ingested data.
- Developed a reusable framework using Python with Google Cloud Functions and Eventarc triggers for file validation. The framework automatically triggers Cloud Workflows when source files are uploaded to Google Cloud Storage, specifically initiating dataflow jobs after validation of 9 source files. This automation streamlines the ETL process and enhances operational efficiency.
- Created and deployed a reusable ETL Framework utilizing Dataflow and Apache Beam for 3 different data sources. The framework efficiently handled data volumes ranging from 70 million records to 100 million records daily, transferring data from GCS to BigQuery.
- Designed a near-real-time pipeline to update retail transaction data from IOT sensors to BigQuery and Cloud SQL using Pub/Sub and Dataflow.
- Developed a Dataflow Pipeline for writing to Cloud SQL [PostgreSQL], incorporating a custom ParDo function to address Apache Beam logic gaps. Optimized Cloud SQL connections using JDBC Libraries, thus avoiding creating multiple connection pools.
- Orchestrated the execution of data pipelines based on both daily and monthly schedules, optimizing data processing and ingestion to align with business requirements and deadlines.
- Created Triggering Mechanisms for front end UI in BigQuery to generate reports from Stored Procedures by utilizing Rest API Calls, Cloud Functions and UDF in BigQuery.

Systems Engineer (Data Engineer)

Sep 2020 – Oct 2022

Tata Consultancy Services (Industry: Telecommunications)

Chennai, India

- Fine-tuned and managed 150+ ETL pipelines on Google Cloud using Cloud Data Fusion, processing 50TB of raw data and transforming into actionable insights, directly impacting business decision-making.
- Managed a 5-member cross-functional (Development, Quality Analyst, Data Architect, support) team and coordinated with six senior business partners toward the successful launch of production releases.
- Analyzed, classified, and assessed the fine-tuning of drivers, memory, and executor cores to optimize Dataproc Spark cluster performance, resulting in a 24% increase in pipeline efficiency. This initiative led to substantial time savings, reducing processing time by 30 hours per week.
- Developing Wrangler directives (Recipes) for data cleansing and transformations which satisfy the customer requirement as well as format data suitable for the resultant system.

Technical Skills

Pub/Sub, Google Dataflow, Google Data Fusion, Google Big Query, Google Data Proc, Google Cloud SQL, Google Cloud Functions, Google Cloud Workflow, Google Cloud Composer, Google Cloud Storage, Google Cloud KMS, Google Secret Manager, Apache Beam, Scala Spark, PySpark, Apache Airflow, Oracle SQL, PostgreSQL, CDAP Directives, Apache Spark, Scala, Hudi, Hive, Python, Github

Professional Certifications

Google Cloud [Associate Cloud Engineer, Professional Data Engineer], Astronomer for Apache Airflow Fundamentals.